

# Sungguk Cha

## Biography

I am Researcher at the Service AI Research Team of LG Uplus closely working with Dr. Mintae Kim and DongWook Kim. Before that, I have spent wonderful times as a Technical Research Personnel for the military duty working with Dr. Cheoljong Yang and Jusung Lee at NCSoft, and a number of fantastic colleagues at MINDs Lab Inc. I obtained my M.S. degree in Ulsan National Institute of Science and Technology.

My research interests cover the broad area of machine learning, multi-modal learning, computer vision, language modeling and quantitative finance. My current position focuses on synthesizing train/evaluate data for (vision-)language models.

## Publications

- Frame Entrepreneurs in an AI Agent Community: Concentrated Identity-Claim Production on Moltbook  
**Sungguk Cha**, DongWook Kim.  
Communication, Information Technologies, and Media Sociology section of the American Sociological Association (CITAMS), 2026.
- ReinPool: Reinforcement Learning Pooling Multi-Vector Embeddings for Retrieval System  
**Sungguk Cha**, DongWook Kim, Mintae Kim, Youngsub Han, Byoung-Ki Jeon, Sangyeob Lee  
Preprint, 2026.
- Annotation-Free Reinforcement Learning Query Rewriting via Verifiable Search Reward  
**Sungguk Cha**, DongWook Kim, Taeseung Hahn, Mintae Kim, Youngsub Han, Byoung-Ki Jeon  
Preprint, 2025.
- Visually Dehallucinative Instruction Generation: Know What You Don't Know  
**Sungguk Cha**, Jusung Lee, Younghyun Lee, Cheoljong Yang.  
Preprint, 2024.
- Visually Dehallucinative Instruction Generation  
**Sungguk Cha**, Jusung Lee, Younghyun Lee, Cheoljong Yang.  
Proceedings of IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2024.
- Question Answer Instruction: Unlocking Multi-Modal Large Language Model to Domain-Specific Visual Multi-Tasks  
Jusung Lee, **Sungguk Cha**, Younghyun Lee, Cheoljong Yang.  
Preprint, 2024.
- Zero-Shot Semantic Segmentation via Spatial and Multi-scale aware Visual Class Embedding  
**Sungguk Cha\***, Yooseung Wang\*.  
*Pattern Recognition Letters (PRL)*, Volume 158, June 2022, Pages 87-93.

## Acknowledgements

- Top Finisher in StarCraft AI Tournaments  
Entry name: Zia Bot
  - StarCraft AI Competitions, Bots and Tournament Manager Software.  
Michal Čertický, David Churchill, Kyung-Joong Kim, Martin Čertický, Richard Kelly.  
IEEE Transactions on Games (ToG): 1-13. DOI: 10.1109/TG.2018.2883499. Print ISSN: 2475-1502. Online ISSN: 2475-1510. 2018.
  - The Current State of StarCraft AI Competitions and Bots.  
M. Čertický, D. Churchill.  
In Proceedings of the AIIDE 2017 Workshop on Artificial Intelligence for Strategy Games. 2017.

## Experiences

LG Uplus

August 2024 - Present

- Service AI Research Team (Jan. 2026 - Present)
  - Reference: Dr. Mintae Kim
  - Multimodal Embedding alignment and Interleaved Data Processing.
  - GUI Agent.
- RAG AI Technical Team (Jan. 2025 - Dec. 2025)
  - Reference: Dr. Mintae Kim
  - Multimodal RAG, document question answering
    - Multimodal RAG framework implementation including document embedding and question answering.
  - Multi-turn RAG agent developments and evaluation dealing with the company secretary documents. (reference Geun Lim)
    - Built the train/evaluate data for the multiturn query rewrite agent.
    - Built the evaluate data and the llm-as-judge model for the multiturn RAG scenario.
    - Closely worked with the related departments preparing benchmarks, data and handling issues related to the RAG system.
  - Document Parsing (reference Soyeon Jo, Heonjin Ha)
    - Advancing document parsing (Document Layout Analysis(DLA), VLM parsing) : analysis, evaluation and synthesizing.
  - Query Rewriting (reference DongWook Kim, Taeseung Hahn)
    - (Data, Index)-agnostic query rewriting approach for retrieval system.
- LLM Technical Team (Aug. 2024 - Dec. 2024)
  - Developed multi-turn RAG data synthesis approaches.
  - MWC 2025 preparation. LLM for digital humanoid implementation.

## OAQ Co. Ltd.

June 2024 - March 2025

- Auditor

## NCSoft

Sep. 2022 - May 2024

- Technical Research Personnel for Alternative Military Service (Sep. 2022 - Feb. 2024)
- Image Generation Team, Multimodal AI Lab. (Feb. 2024 - May 2024)
  - VQA (VLM) research and development for digital humans.
  - Multi-modal model research and development for game-domain. It includes multi-modal crawling webscale data.
  - Scale-up Data Curation for Generative Foundation Models: Replaced human-annotation in captioning (tagging) and resulted in dramatic improvements in developing image generative models.
- Recognition Team, Vision AI Lab., NC Research (Sep. 2022 - Feb. 2024)
  - Multimodal Recognition AI: Generated multi-modal data for novel tasks. Researched on evaluating large multimodal models. Responsible and dehallucinative multimodal data generation.
  - Conversational AI
  - Facial Expression Recognition

## MINDs Lab., Inc.

Sep. 2020 - Sep. 2022

- The company now goes by the name maum.ai.
- Technical Research Personnel for Alternative Military Service (Feb. 2021 - Sep. 2022)
- Senior AI Scientist (Mar. 2021 - Sep. 2022)
  - Co-led the research team (HR, managing, etc.)

- Developed and implemented fundamental visual recognition applications, focusing on classification such as firearms and OCR, detection including mover objects and car damage, segmentation with a specialization in fashion items, and person re-identification for tracking corona-infected individuals.
- Contributed to Conversational AI, specializing in human-object interaction through achievements in Visual Question Answering, ATM abnormal behavior detection, and human-object detection/classification.

## Statistical Artificial Intelligence Lab

Jul. 2018 - Feb. 2019

- <http://sailab.kaist.ac.kr/>
- Adviser: Prof. Jaesik Choi
- Undergraduate Researcher
- Research Topic: Drivable Area Semantic Segmentation

## Agents and Robotic Transportation Lab

Dec. 2017 - Feb. 2018

- Adviser: Prof. Tsz-Chiu Au
- Undergraduate Researcher
- Research Topic: StarCraft AI Algorithm

## Machine Learning & Artificial Intelligence

Sep. 2017 - Dec 2017

- <https://www.mlai-kaist.com/>
- Adviser: Prof. Sung Ju Hwang
- Undergraduate Researcher
- Research Topic: Variational Auto Encoder for Text

## SKT T-Brain

Mar. 2017

- Research service invited by Hyunsoo Kim (Superb AI CEO)
- Passed interviews
- Research Topic: Machine Learning Application for StarCraft

## Professional Services

### Conference Talks:

- Frame Entrepreneurs in an AI Agent Community: Concentrated Identity-Claim Production on Moltbook  
**Sungguk Cha**, DongWook Kim.  
Communication, Information Technologies, and Media Sociology section of the American Sociological Association (CITAMS), 2026.
- 2026 American Sociological Association (ASA) Annual Meeting CITAMS Roundtable: "The Synthetic Social Graph: Emergent Behavior in AI Agent Communities"
- LG Software Developer Conference (LG SDC) 2025.

### Conference Reviewer:

- Association for the Advancement of Artificial Intelligence (AAAI) 2026.
- IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) 2025, 2026.
- IEEE International Joint Conference on Neural Networks (IJCNN) 2025, 2026.

## Educations

### Ulsan National Institute of Science and Technology

Mar. 2019 - Feb. 2021

- **Master of Science** in Computer Science and Engineering (GPA 3.94/4.30)
- Thesis: Zero-Shot Semantic Segmentation via Visual Class Embedding
- Full Merit based Scholarship

## Ulsan National Institute of Science and Technology

Mar. 2016 - Feb. 2019  
Mar. 2014 - Feb. 2015

- **Bachelor of Science** in
  - Computer Science and Engineering (GPA 3.74/4.30)
  - Electrical Engineering
- **Cum Laude**
- Full Merit based Scholarship

## Honors & Awards

- Dacon AI Bit Trader Competition **29th (Top 21%)** Mar 2021
- NAVER LABS Mapping & Localization Challenge 8th place Jun 2020
- Kaggle/iMaterialist (Fashion) 2019 at FGVC6/fashion item panoptic segmentation/(29th/194 (Top 15%))
- Kaggle/Google Analytics Customer Revenue Prediction/(570th/3611 (Top16%))
- Student StarCraft AI Tournament **4th** Feb 2018
- Naver UNIST Undergraduate Poster Award, **Bronze** Prize Dec 2017  
Topic: Reinforcement Learning in StarCraft in Real-Time using Deep Learning
- Samsung Dream Class Mentoring Scholarship Spring 2017
- AAAI AIIDE StarCraft AI Competition, 28th 2017
- IEEE CIG StarCraft AI Competition **3rd** 2017
- Student StarCraft AI Tournament **3rd** Feb 2017
- AAAI AIIDE StarCraft AI Competition, 11th 2016
- IEEE CIG StarCraft AI Competition 10th 2016
- Korean Olympiad of Informatics **Encouragement** Prize 2013  
Regional representative of Daejeon
- Daejeon Olympiad of Informatics **Gold Prize** 2013
- Korean Olympiad of Informatics **Encouragement** Prize 2012  
Regional representative of Daejeon
- Daejeon Olympiad of Informatics **Gold Prize** 2012
- Korean Olympiad of Informatics **Encouragement** Prize 2010  
Regional representative of Daejeon
- Daejeon Olympiad of Informatics **Gold Prize** 2010

## Teaching

- [TA] UNIST/CSE241: Object Oriented Programming Spring 2020
- [TA] UNIST/CSE251: System Programming Fall 2019
- **[Tutor] PyCon Korea** Aug 2019  
Topic: Explainable AI
- [TA] UNIST/ITP117: Introduction to AI Programming II Spring 2019
- [TA] UNIST/AHS161: What is "I" Spring, Fall 2017

## Skills

- Languages
  - **Korean** | **English** | **Python** | C/C++ | MFC

- Frameworks
  - Deep learning: **PyTorch, Multimodal, RLHF/RLAIF, RAG agents**
  - Collaboration: **ATLASSIAN**, Tigris
  - Developments: **Data Curation, Interleaved Data Processing**, Git, Docker, gRPC